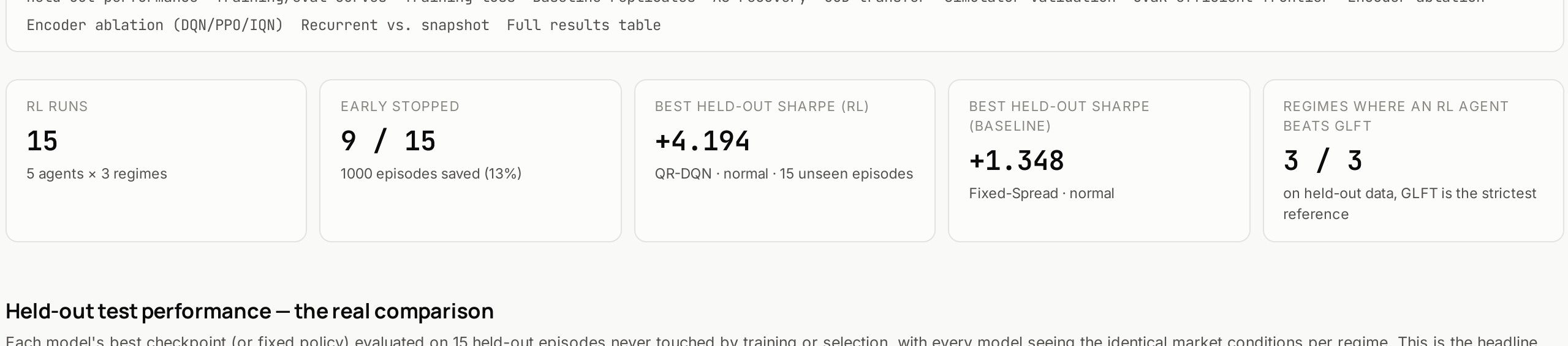


DistRL Market Maker – Training & Evaluation Results

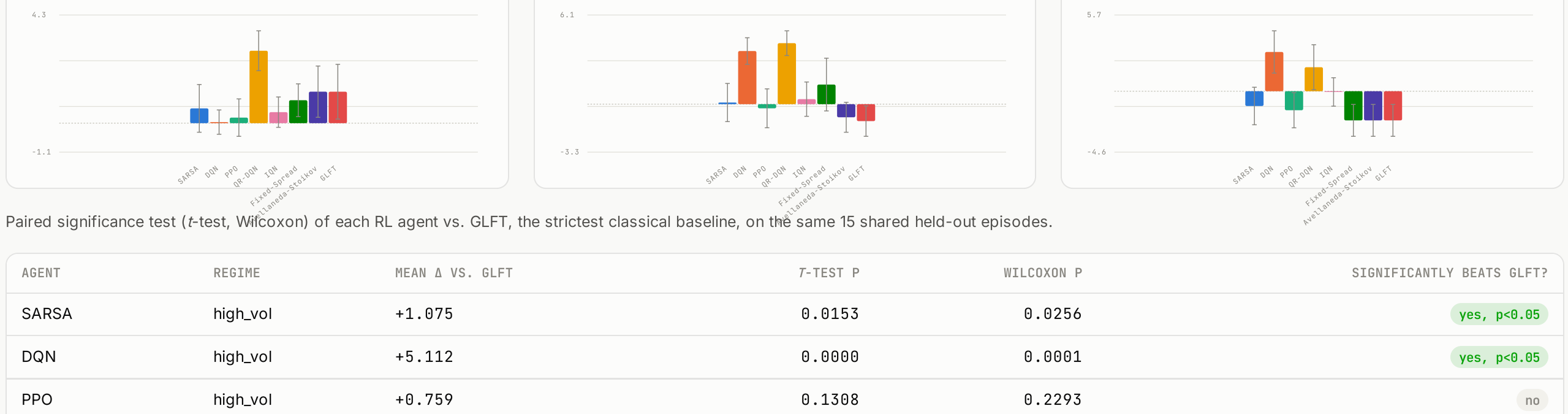
seed=42 budget=500 ep encoder=handcrafted reward=asymmetric

5 RL agents vs. 3 analytical baselines across 3 volatility regimes in an ABIDES-Gym limit-order-book simulator, plus out-of-distribution transfer, CVarR risk-sensitivity, and state-representation ablations.



Held-out test performance – the real comparison

Each model's best checkpoint (or fixed policy) evaluated on 15 held-out episodes never touched by training or selection, with every model seeing the identical market conditions per regime. This is the headline result – everything below is a training-time diagnostic.

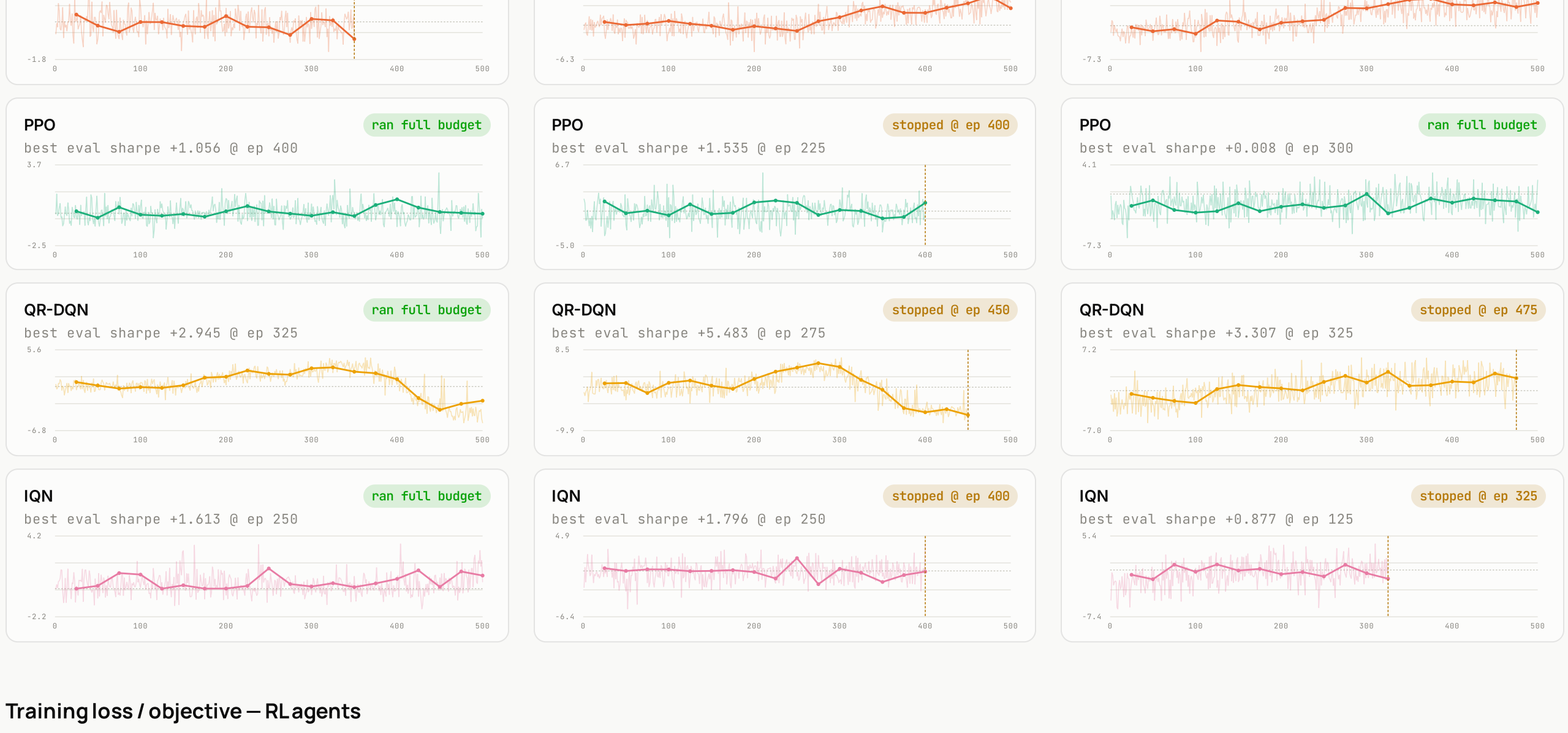


Paired significance test (t-test, Wilcoxon) of each RL agent vs. GLFT, the strictest classical baseline, on the same 15 shared held-out episodes.

AGENT	REGIME	MEAN & VS. GLFT	T-TEST P	WILCOXON P	SIGNIFICANTLY BEATS GLFT?
SARSA	high_vol	+1.075	0.0153	0.0256	yes, p<0.05
DQN	high_vol	+5.112	0.0000	0.0001	yes, p<0.05
PPO	high_vol	+0.797	0.1308	0.2293	no
QR-DQN	high_vol	+3.921	0.0000	0.0001	yes, p<0.05
IQN	high_vol	+2.125	0.0001	0.0002	yes, p<0.05
SARSA	low_vol	-0.657	0.0799	0.0946	no
DQN	low_vol	-1.199	0.0029	0.0034	no
PPO	low_vol	-1.021	0.0229	0.0215	no
QR-DQN	low_vol	+1.613	0.0000	0.0001	yes, p<0.05
IQN	low_vol	-0.806	0.0036	0.0043	no
SARSA	normal	+1.278	0.0003	0.0001	yes, p<0.05
DQN	normal	+4.818	0.0000	0.0001	yes, p<0.05
PPO	normal	+0.880	0.0614	0.0151	yes, p<0.05
QR-DQN	normal	+5.358	0.0000	0.0001	yes, p<0.05
IQN	normal	+1.510	0.0030	0.0015	yes, p<0.05

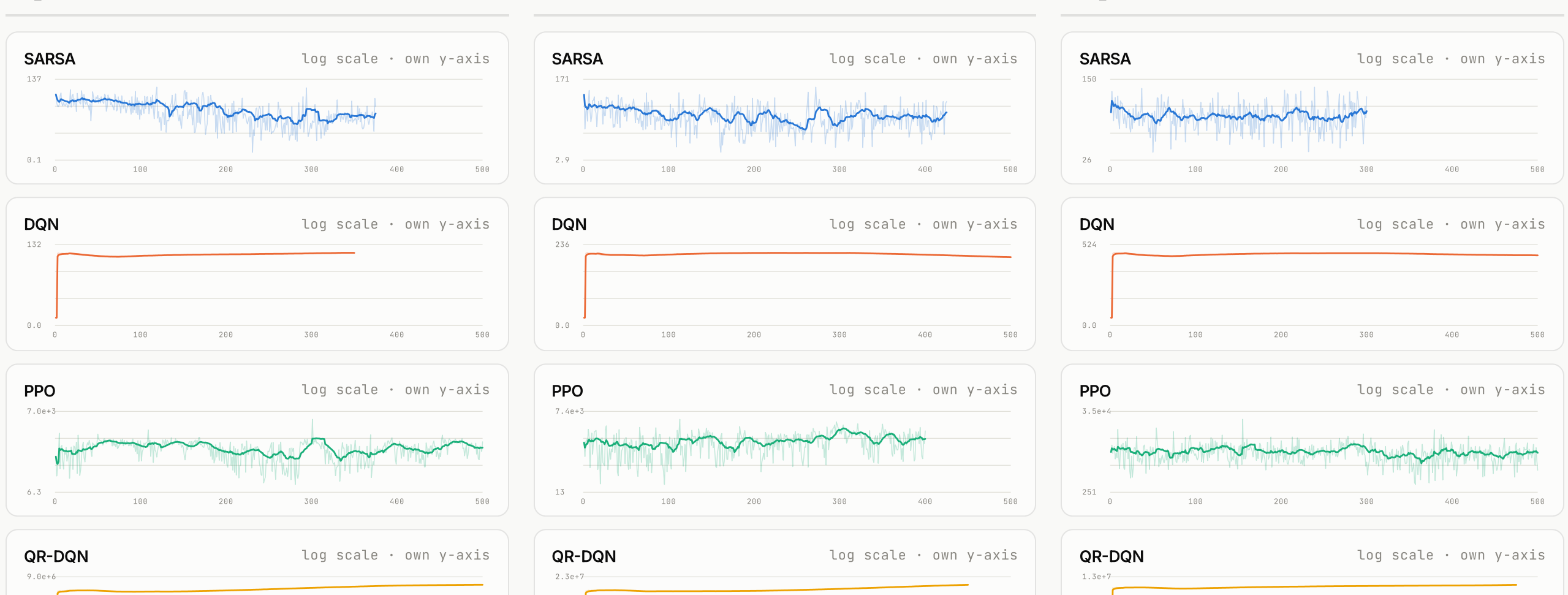
Training & evaluation curves – RL agents

Raw per-episode training Sharpe vs. smoothed eval Sharpe, the signal early stopping watches, for each RL agent per regime.



Training loss / objective – RL agents

Each agent's own training-objective loss on a log scale – not comparable across algorithms, since each optimizes a different loss function.



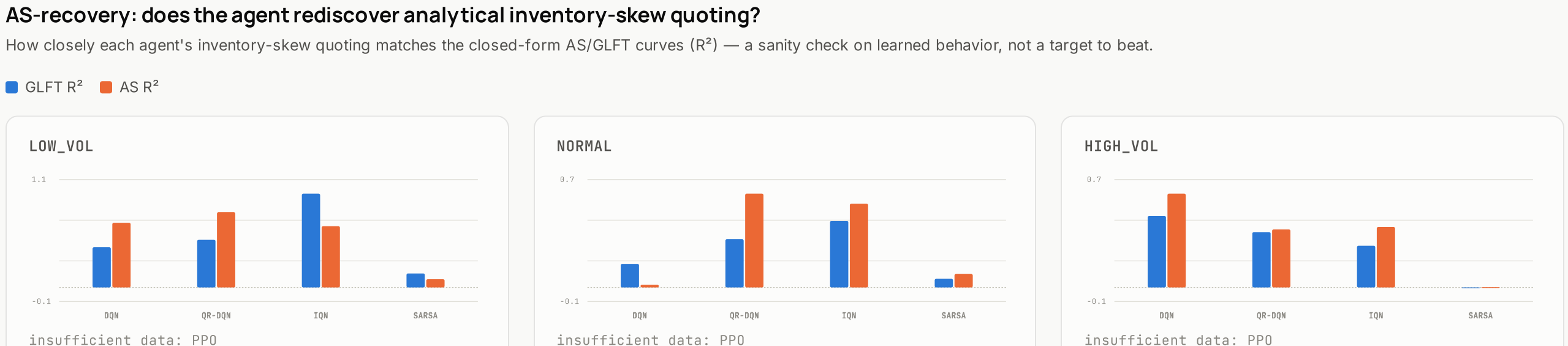
Baseline replicate performance

Performance variance across 50 independent replicates of each fixed (non-learning) baseline policy, per regime.



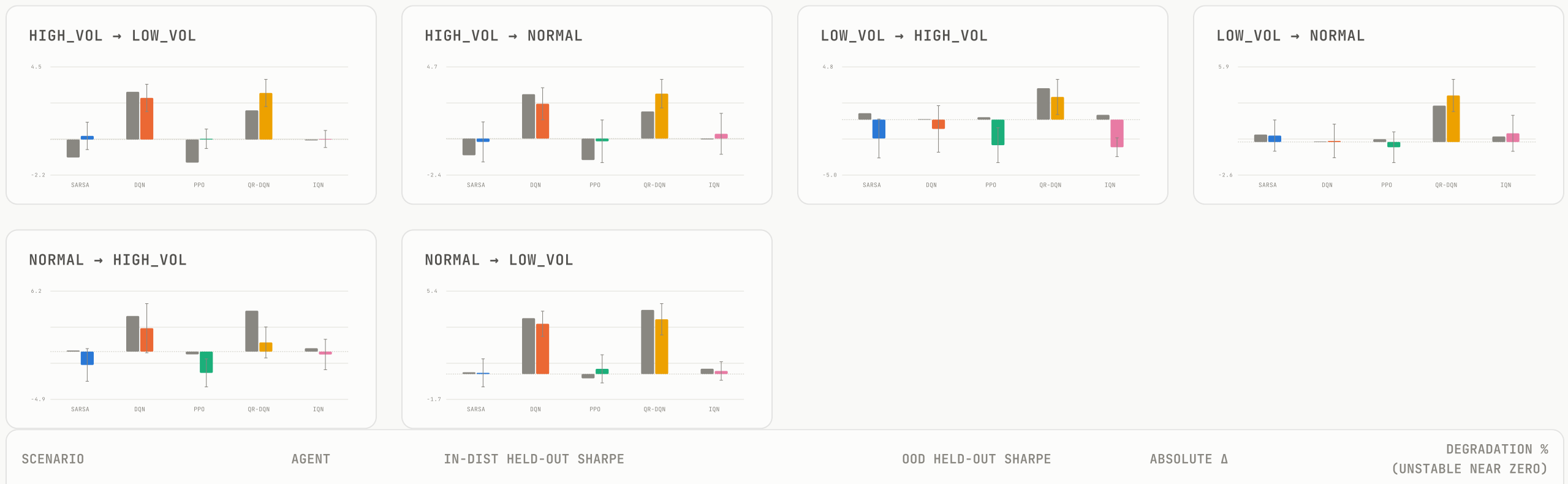
AS-recovery: does the agent rediscover analytical inventory-skew quoting?

How closely each agent's inventory-skew quoting matches the closed-form AS/GLFT curves (R²) – a sanity check on learned behavior, not a target to beat.



OOD transfer: agents evaluated outside their training regime

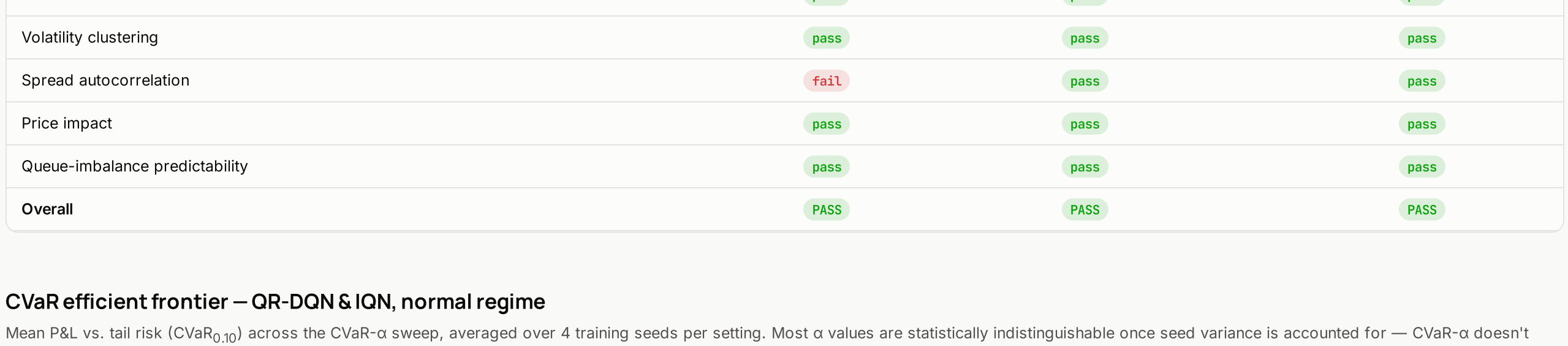
A random-policy audit of the simulator asset against 5 standard empirical market-microstructure regularities, independent of any trained agent – validates that the environment produces realistic dynamics to learn from.



SCENARIO	AGENT	IN-DIST	HELD-OUT SHARPE	OOD HELD-OUT SHARPE	ABSOLUTE Δ	DEGRADATION % (UNSTABLE: NEAR ZERO)
high_vol → low_vol	DQN	+2.933		+2.559	-0.374	+13%
high_vol → low_vol	IQN	-0.054		+0.034	-0.088	-163%
high_vol → low_vol	QR-DQN	+1.793		+2.862	+1.069	+60%
high_vol → low_vol	SARSA	-1.104		+0.224	+1.328	-120%
high_vol → low_vol	PPO	-1.419		+0.846	+1.465	-103%
high_vol → normal	DQN	+2.933		+2.300	-0.633	+22%
high_vol → normal	IQN	-0.054		+0.313	+0.368	-678%
high_vol → normal	SARSA	-1.104		-0.220	+0.884	-80%
high_vol → normal	QR-DQN	+1.793		+2.970	+1.178	-66%
high_vol → normal	PPO	-1.419		-0.181	+1.238	-87%
low_vol → high_vol	IQN	+0.436		-2.505	-2.939	+677%
low_vol → high_vol	PPO	+0.219		-2.320	-2.539	+1159%
low_vol → high_vol	SARSA	+0.593		-1.710	-2.293	+393%
low_vol → high_vol	DQN	+0.041		-0.845	-0.886	n/a (in-dist = 0)
low_vol → high_vol	QR-DQN	+2.853		+0.059	-0.794	+28%
low_vol → normal	PPO	+0.219		-0.411	-0.630	+288%
low_vol → normal	SARSA	+0.593		+0.502	-0.081	+14%
low_vol → normal	DQN	+0.041		+0.082	+0.041	n/a (in-dist = 0)
low_vol → normal	IQN	+0.436		+0.681	+0.247	-57%
low_vol → normal	QR-DQN	+2.853		+3.651	+0.798	-28%
normal → high_vol	QR-DQN	+4.194		+0.939	-3.255	+78%
normal → high_vol	PPO	-0.284		-2.184	-1.900	+668%
normal → high_vol	SARSA	+0.114		-1.370	-1.484	+1307%
normal → high_vol	DQN	+3.653		+2.400	-1.253	+34%
normal → low_vol	IQN	+0.345		-0.290	-0.643	+16%
normal → low_vol	QR-DQN	+4.194		+3.581	-0.613	+15%
normal → low_vol	DQN	+3.653		+3.286	-0.367	+10%
normal → low_vol	IQN	+0.345		+0.197	-0.148	+43%
normal → low_vol	SARSA	+0.114		+0.075	-0.039	+34%
normal → low_vol	PPO	-0.284		+0.338	+0.622	-219%

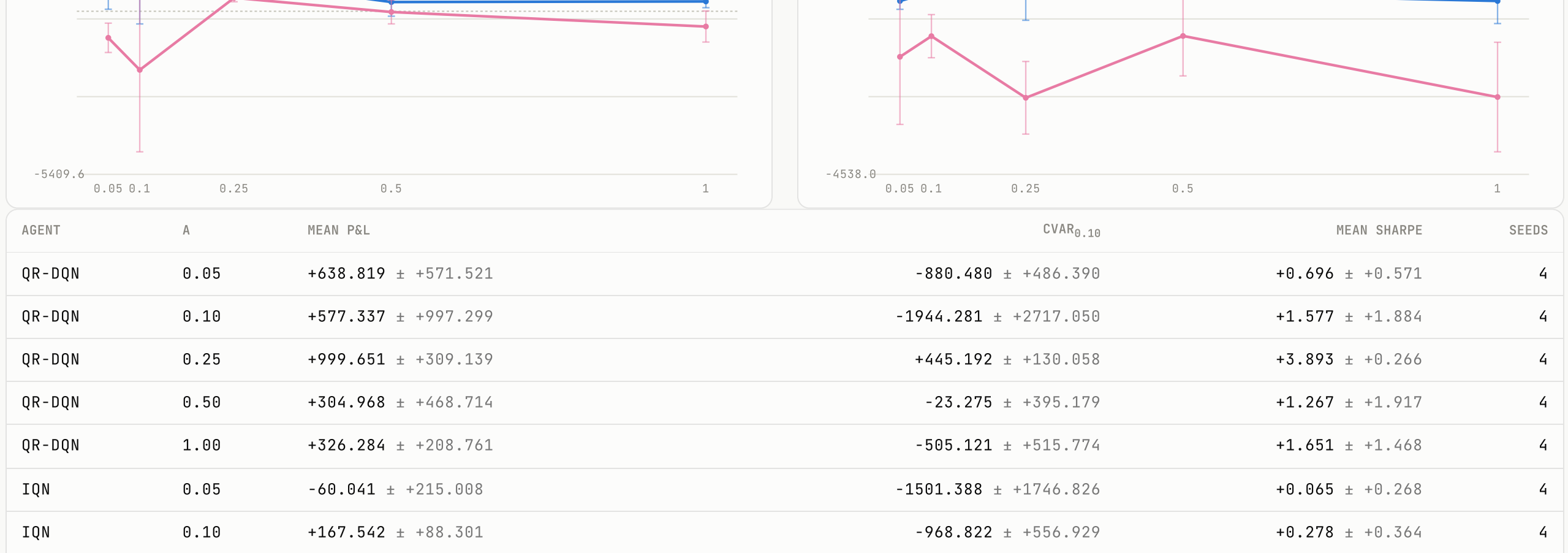
Simulator validation – stylized facts

A random-policy audit of the simulator asset against 5 standard empirical market-microstructure regularities, independent of any trained agent – validates that the environment produces realistic dynamics to learn from.



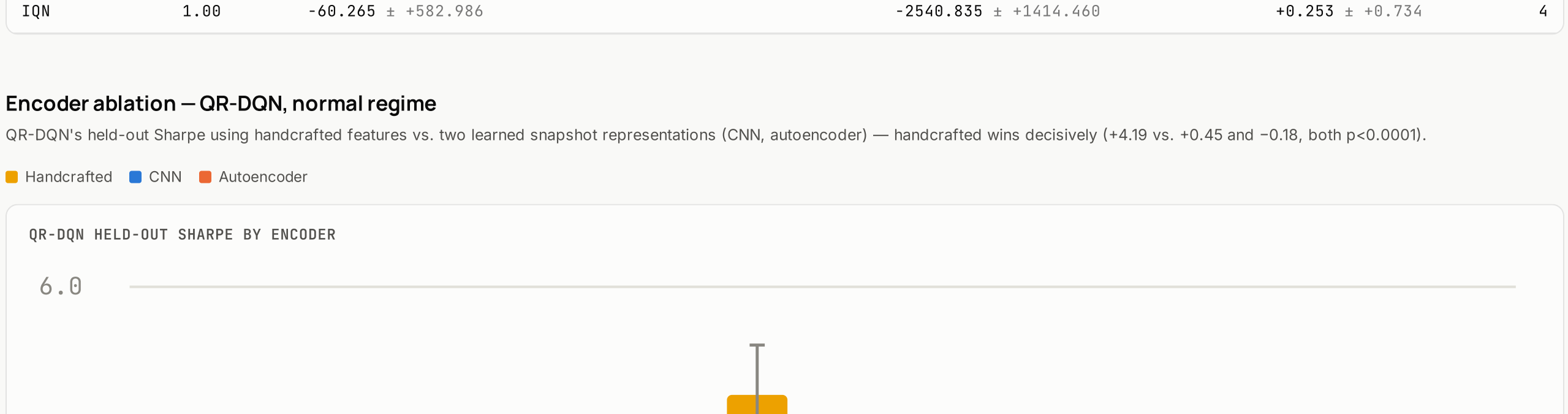
CVarR efficient frontier – QR-DQN & IQN, normal regime

Mean P&L vs. tail risk (CVarR_{0.99}) across the CVarR-α sweep, averaged over 4 training seeds per setting. Most α values are statistically indistinguishable once seed variance is accounted for – CVarR-α doesn't produce the smooth risk/return tradeoff the method is meant to deliver at this training budget.



Encoder ablation – QR-DQN, normal regime

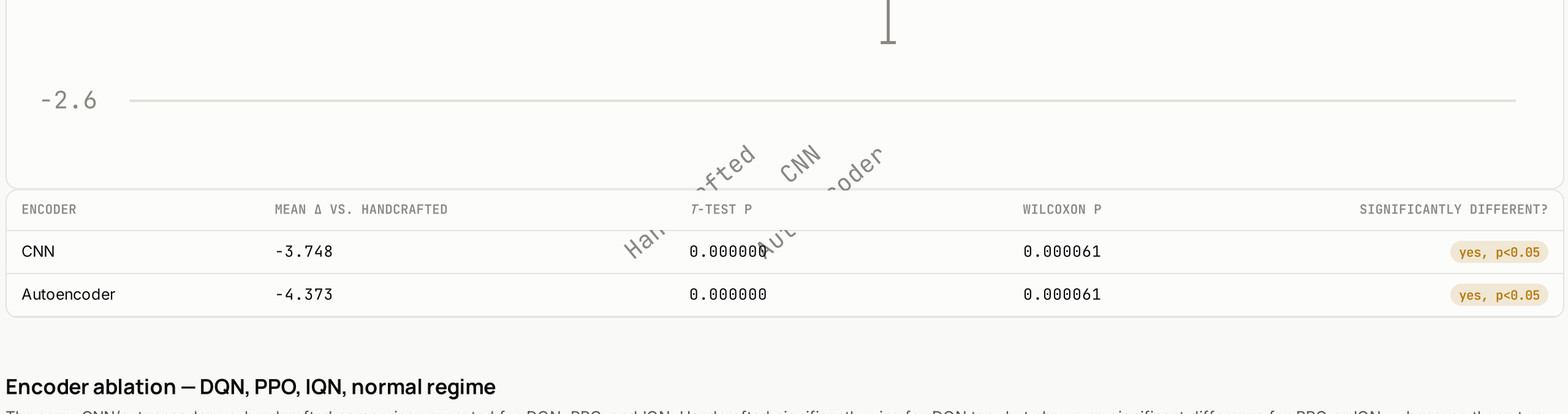
QR-DQN's held-out Sharpe using normal features vs. two learned snapshot representations (CNN, autoencoder) – handcrafted wins decisively (+4.19 vs. +0.45 and -0.18, both p<0.0001).



ENCODER	MEAN Δ VS. HANDCRAFTED	T-TEST P	WILCOXON P	SIGNIFICANTLY DIFFERENT?
CNN	-3.748	0.000000	0.000061	yes, p<0.05
Autoencoder	-4.373	0.000000	0.000061	yes, p<0.05

Encoder ablation – DQN, PPO, IQN, normal regime

The same CNN/autoencoder-vs-handcrafted comparison repeated for DQN, PPO, and IQN. Handcrafted significantly wins for DQN too, but shows no significant difference for PPO or IQN – because those two agents' own handcrafted baselines are themselves weak, not because their learned encoders work better.

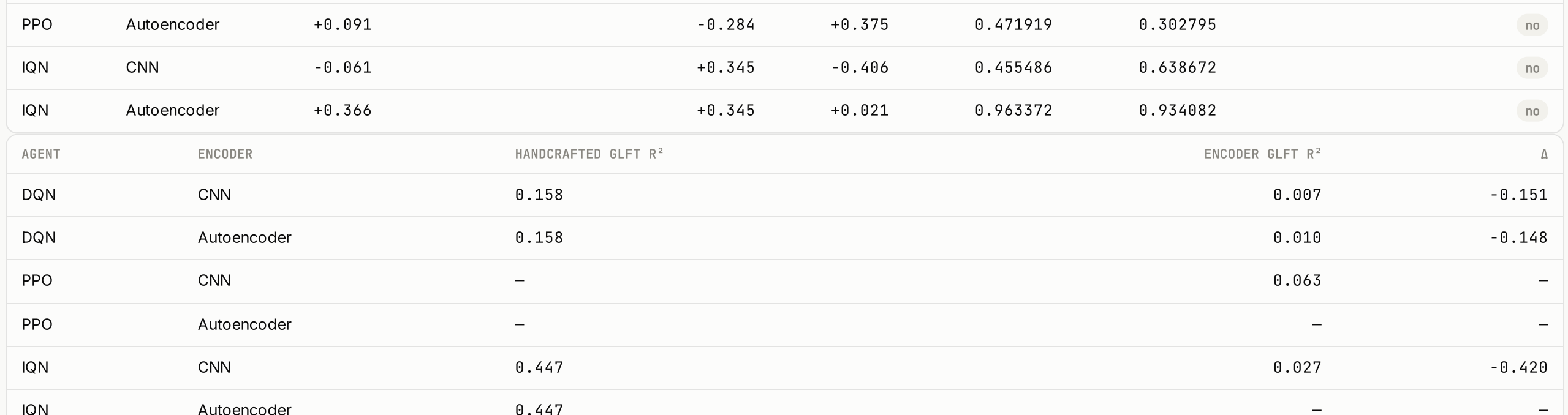


AGENT	ENCODER	ENCODER SHARPE	HANDCRAFTED SHARPE	MEAN Δ	T-TEST P	WILCOXON P	SIGNIFICANTLY DIFFERENT?
DQN	CNN	+0.376	+3.653	-4.049	0.000000	0.000061	yes, p<0.05
DQN	Autoencoder	+0.020	+3.653	-3.633	0.000000	0.000061	yes, p<0.05
PPO	CNN	+0.466	-0.284	+0.750	0.100613	0.120544	no
PPO	Autoencoder	+0.091	-0.284	+0.375	0.471919	0.302795	no
IQN	CNN	-0.061	+0.345	+0.406	0.455406	0.638672	no
IQN	Autoencoder	+0.366	+0.345	+0.021	0.963372	0.934082	no

AGENT	ENCODER	ENCODER SHARPE	HANDCRAFTED GLFT R²	ENCODER GLFT R²	Δ
DQN	CNN	0.158	0.158	0.007	-0.151
DQN	Autoencoder	0.158	0.158	0.007	-0.148
PPO	CNN	—	—	0.063	—
PPO	Autoencoder	—	—	—	—
IQN	CNN	0.447	0.447	0.027	-0.420
IQN	Autoencoder	0.447	0.447	—	—

Recurrent vs. snapshot: does temporal memory help?

Each agent's standard snapshot-encoder policy vs. the same agent with an LSTM backbone in its place, across 10 agent/regime/CVarR-risk combinations. Recurrent integration never significantly helps, significantly hurts QR-DQN and DQN in the normal regime, and its effect when combined with CVarR risk-sensitivity is alpha-dependent rather than a consistent interaction.



VARIANT	SNAPSHOT SHARPE	RECURRENT SHARPE	MEAN Δ	T-TEST P	WILCOXON P	SIGNIFICANTLY DIFFERENT?
QR-DQN (low_vol)	+2.853	+3.087	+0.234	0.441011	0.561401	no
QR-DQN (normal)	+4.194	+1.339	-2.855	0.000077	0.000122	yes, p<0.05
QR-DQN (high_vol)	+1.793	+0.644	-1.149	0.089223	0.276855	no
QR-DQN (α=0.05)	-0.050	+0.227	+0.286	0.241471	0.276855	no
QR-DQN (α=0.10)	+3.260	+0.565	-2.695	0.000003	0.000061	yes, p<0.05
QR-DQN (α=0.50)	-0.930	+2.287	+3.217	0.000002	0.000122	yes, p<0.05
QR-DQN (α=1.00)	-0.101	+0.538	+0.639	0.229020	0.229009	no
DQN (normal)	+3.653	+0.346	-3.307	0.000000	0.000061	yes, p<0.05
IQN (normal)	+0.345	+0.014	-0.331	0.458056	0.561401	no
PPO (normal)	-0.284	-0.236	+0.049	0.920456	0.638672	no

VARIANT	SNAPSHOT GLFT R²	RECURRENT GLFT R²	Δ
QR-DQN (low_vol)	—	0.046	—
QR-DQN (normal)	—	0.024	—
QR-DQN (high_vol)	—	0.173	—
QR-DQN (α=0.05)	—	0.522	—
QR-DQN (α=0.10)	—	0.700	—
QR-DQN (α=0.50)	—	0.245	—
QR-DQN (α=1.00)	—	0.780	—
DQN (normal)	—	0.452	—
IQN (normal)	—	0.125	—

Full results table

Every figure charted above, in one place.

Full results table												
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AGENT	TYPE	REGIME	EPISODES	STATUS	HELD-OUT SHARPE (15 UNSEEN EP. PAID)	± STD	PEAK EVAL SHARPE	EP	FINAL EVAL SHARPE (LAST 5, 30-SAMPLE)	TRAIN-TEST MEAN (DIAGNOSTIC ONLY)	GLFT R ²	AS R ²
SARSA	RL	low_vol	375 / 500	early stopped	+0.583	0.94	+0.992	225	-0.248	+0.149	0.146	0.886
SARSA	RL	normal	425 / 500	early stopped	+0.114	1.31	+2.103	225	-0.777	+0.044	0.058	0.898
SARSA	RL	high_vol	300 / 500	early stopped	-1.184	1.48	-0.129	200	-1.108	-1.346	0.080	0.802
DQN	RL	low_vol	350 / 500	early stopped	+0.041	0.48	+0.356	25	-0.289	-0.080	0.421	0.877
DQN	RL	normal	500 / 500	complete	+3.653	0.91	+5.327	475	+4.147	+0.993	0.158	0.818
DQN	RL	high_vol	500 / 500	complete	+2.933	1.58	+5.772	375	+4.633	+1.751	0.449	0.589
PPO	RL	low_vol	500 / 500	complete	+0.219	0.74	+1.056	400	+0.830	+0.038	-	-
PPO	RL	normal	400 / 500	early stopped	-0.284	1.34	+1.535	225	-0.229	-0.132	-	-
PPO	RL	high_vol	500 / 500	complete	-1.419	1.31	+0.008	300	-1.497	-1.275	-	-
QR-DQN	RL	low_vol	500 / 500	complete	+2.853	0.79	+2.945	325	-2.810	+0.429	0.080	0.788
QR-DQN	RL	normal	450 / 500	early stopped	+4.194	0.85	+5.483	275	-5.703	+1.003	0.324	0.636
QR-DQN	RL	high_vol	475 / 500	early stopped	+1.793	1.68	+3.307	325	+2.224	+0.328	0.347	0.364
IQN	RL	low_vol	500 / 500	complete	+0.434	0.68	+1.613	250	+0.854	+0.364	0.983	0.641
IQN	RL	normal	400 / 500	early stopped	+0.345	1.18	+1.796	250	-0.741	+0.117	0.447	0.563
IQN	RL	high_vol	325 / 500	early stopped	-0.054	1.06	+0.877	125	-0.357	-0.618	0.262	0.379
Fixed-Spread	baseline	low_vol	50 / 50	n/a (L1d.)	+0.904	0.65	-	-	-	+0.959	-	-